

LAB # 02

WORKING WITH HTML5

OBEJCTIVE:

What is New in HTML5?

HTML 5 Introduction

HTML5 is a markup language used for structuring and presenting content on the World Wide Web. It is the fifth and current major version of the HTML standard.

DOCTYPE

The **DOCTYPE** declaration for HTML5 is very simple:

```
<!DOCTYPE html>
```

The **document type declaration (DOCTYPE)** is *required* in HTML5 documents so that browsers render the page in **standards mode**, according to the HTML and CSS specifications.

Meta Elements

The character encoding (charset) declaration is also very simple:

```
<meta charset="UTF-8">
```

Search engines catalog sites by following links from page to page and saving identification and classification information for each page. One way that search engines catalog pages is by reading the content in each page's meta elements, which specify information about a document.

EXAMPLE

```
<!-- <meta> tags provide search engines with -->
<!-- information used to catalog a site      -->
<meta name = "keywords" content = "web page, design,
    HTML5, tutorial, personal, help, index, form,
    contact, feedback, list, links, deitel">
<meta name = "description" content = "This website will
    help you learn the basics of HTML5 and web page design
    through the use of interactive examples and
    instruction.">
```

Two important attributes of the meta element are:

name: which identifies the type of meta element.

content: which provides information search engines use to catalog pages.

The content attribute of a keywords meta element provides search engines with a list of words that describe the page. These words are compared with words in search requests.

The content attribute of a description meta element provides a three- to four-line description of a site, written in sentence form. Search engines also use this description to catalog your site and sometimes display this information as part of the search results.

New HTML5 Elements

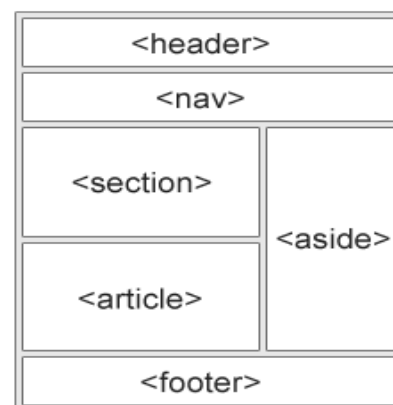
The most interesting new HTML5 elements are:

1. New **semantic elements** like <header>, <footer>, <article>, <section>.
2. New **attributes of form elements** like number, date, time, calendar, and range.
3. New **graphic elements**: <svg> and <canvas>.
4. New **multimedia elements**: <audio> and <video>.

SEMANTIC ELEMENTS

A semantic element clearly describes its meaning to both the browser and the developer. Clearly defines its content. HTML5 offers new semantic elements to define different parts of a web page:

- <article>
- <aside>
- <details>
- <figcaption>
- <figure>
- <footer>
- <header>
- <main>
- <mark>
- <nav>
- <section>
- <summary>
- <time>



section Element

- describes a section of a document, usually with a heading for each section.
- section elements can be nested.

Example

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<meta charset="utf-8">
```

```
<title>How to define a section in a  
document</title>
```

```
</head>
```

```
<body>
```

```
<section>
```

```
<h1>HTML5</h1>
```

```
<p>HTML5 is the fifth revision of HTML, a  
markup language to present and structure web  
document.</p>  
</section>
```

```
<section>
```

```
<h1>History of HTML5</h1>
```

```
<p>The parent concern of HTML5, The Web  
Hypertext Application Technology Working  
Group (WHATWG) along with W3C began work on  
the new standard in 2004.</p>  
</section>
```

```
</body>
```

```
</html>
```

article element

- describes content that's separate from the main content of the page and might be used or distributed elsewhere, such as a news article, forum post or blog entry.
- article elements can be nested.

EXAMPLE

```
<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8">
  <title>Create an HTML document which uses article element</title>

</head>
<body>

<article>
<h1>Tutorial of HTML5 article element</h1>
<p>HTML5 article element represents independent item like a blog entry in an web
document.</p>
</article>

</body>
</html>
```

MULTIMEDIA ELEMENTS

audio element

The HTML5 <audio> element specifies a standard way to embed audio in a web page. To play an audio file in HTML, use the **<audio>** element:

Syntax

```
<audio controls>
  <source src="horse.ogg" type="audio/ogg">
  <source src="horse.mp3" type="audio/mpeg">
</audio>
```

HTML Audio - How It Works

The **controls** attribute adds audio controls, like play, pause, and volume.

The **<source>** element allows you to specify alternative audio files which the browser may choose from. The browser will use the first recognized format.

The **src** attribute specifies the location of the audio clip.

Example (How to embed audio in a HTML document?)

```
<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8">
  <title>How to add controls to an audio</title>
</head>
<body>

<audio controls>
  <source src="horse.ogg" type="audio/ogg">
  <source src="horse.mp3" type="audio/mpeg">
</audio>

</body>
</html>
```

How to auto play an audio?

```
<audio controls="controls" autoplay>
```

How to play an audio repeatedly?

```
<audio controls loop>
```

How to mute an audio?

```
<audio controls muted>
```

video element

The <video> tag specifies video, such as a movie clip or other video streams.

Currently, there are 3 supported video formats for the <video> element: MP4, WebM, and Ogg.

Syntax

```
<video width="320" height="240" controls>
  <source src="movie.mp4" type="video/mp4">
  <source src="movie.ogg" type="video/ogg">
</video>
```

Attribute	Value	Description
<u>autoplay</u>	autoplay	Specifies that the video will start playing as soon as it is ready
<u>controls</u>	controls	Specifies that video controls should be displayed (such as a play/pause button etc).
<u>height</u>	<i>pixels</i>	Sets the height of the video player
<u>loop</u>	loop	Specifies that the video will start over again, every time it is finished
<u>muted</u>	muted	Specifies that the audio output of the video should be muted
<u>poster</u>	URL	Specifies an image to be shown while the video is downloading, or until the user hits the play button
<u>preload</u>	auto metadata none	Specifies if and how the author thinks the video should be loaded when the page loads
<u>src</u>	URL	Specifies the URL of the video file
<u>width</u>	<i>pixels</i>	Sets the width of the video player

Example (Execute the following code):

```

<!DOCTYPE html>
<html>
<body>

<video width="400" controls>
  <source src="mov_bbb.mp4" type="video/mp4">
  <source src="mov_bbb.ogv" type="video/ogg">
</video>

<p>
Video courtesy of
<a href="https://www.bigbuckbunny.org/" target="_blank">Big Buck Bunny</a>.
</p>

</body>

```

</html>

LAB TASKS

TASK # 1

Create HTML5 document that displays the following:

Famous Cities

London

London is the capital city of England. It is the most populous city in the United Kingdom, with a metropolitan area of over 13 million inhabitants.

Paris

Paris is the capital and most populous city of France.

Tokyo

Tokyo is the capital of Japan, the center of the Greater Tokyo Area, and the most populous metropolitan area in the world.

Note: Use tags like: article, section

TASK # 2

A local university has asked you to create an HTML5 document that allows prospective college students to provide feedback about their campus visit. Your HTML5 document should contain a form with text fields for a name and e-mail.

Provide checkboxes that allow prospective students to indicate what they liked most about the campus. The checkboxes should include: campus, students, location, atmosphere, dorm rooms and sports.

Also, provide radio buttons that ask the prospective students how they became interested in the college.

Options should include: friends, television, Internet and other. In addition, provide a text area for additional comments, a submit button and a reset button. Use post to send the information in the form to <http://www.bahria.com>.

TASK # 3

Create HTML5 page for kindergarten students. It should contain the following:

- Header and footer on the page
- The page must be divided into two sections. First section contains the audio for national anthem. Second section contains at least five videos of famous rhymes.
- It should contain an *image* as an *e-mail link*. Use attribute alt to provide a description of the image and link.
- Include the figures with captions (use **fig caption**).